

- Artificial Intelligence in Industry
 - Project Work (3 CFU)

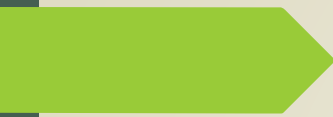
A.Y. 2024/25

Christian Di Buò
Simone Reale

Sustainable City via Trustworthy Digital Twin: a Use Case

AI and GIS for Smarter Urban Green Planning





Background

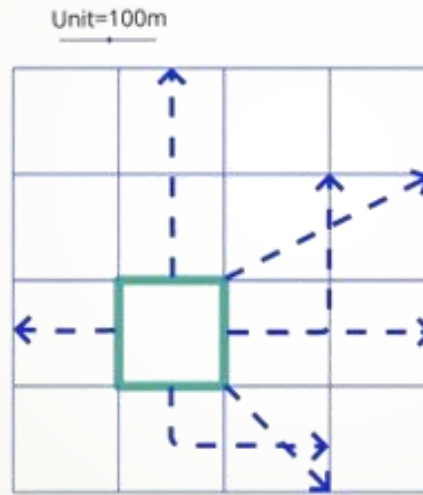


- **TALEA** is funded under the *European Urban Initiative – Innovative Actions (EUI-IA)*, specifically the Greening Cities call.
- Bologna faces urgent threats from urban heat islands (UHI) and urban heat waves (UHW), which require integrated green infrastructure solutions.
- These challenges have environmental, social, economic, and technological dimensions.
- The objective is to promote the creation of a green infrastructure through a system of **TALEA Green Cells (TGCs)**

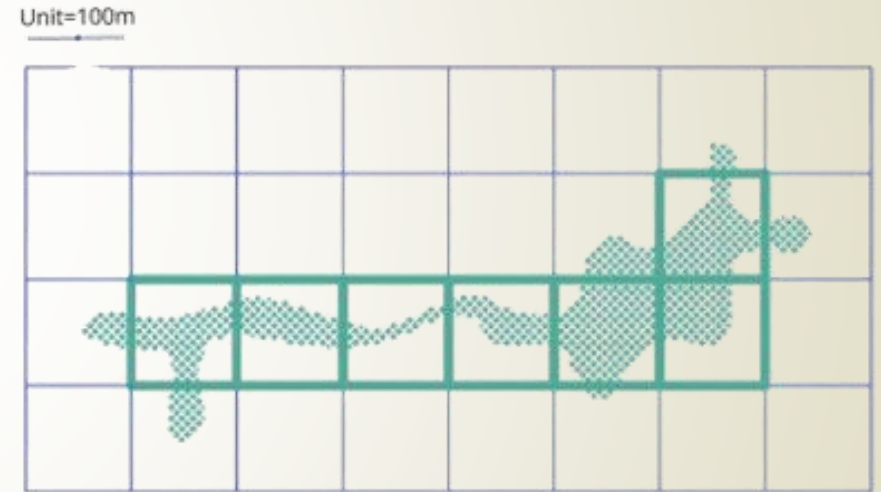
- ▶ TGCs are thought to be 100x100 meters cell designed to create open, accessible “climate refuges”, instead of isolated hotspots, interconnected across the city

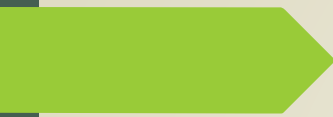


Green Cell basic
measurable unit
(100x100 m)



Green Cell/s sprawl rationale within the 100x100m grid





Study description

- The objective is to find the best areas where to insert new green cells.
- The idea is to divide the city of Bologna into a **grid of 100x100 meter cells** and develop an optimization model which outputs the best n cells.

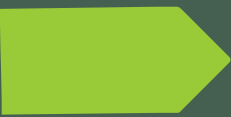
Organization

- Data collection
- Data processing
- Model development
- Parameter experimentations
- Results analysis

Engineering Approach

- ▶ To avoid being dependent from third part geospatial manipulation software, such as QGIS, it has been done the following:
 - A simple **Graphical User Interface (GUI)** was developed. <https://talea-gui.netlify.app/>
Since geospatial data without visualization lose much of their informative value, a GUI was created to display them without requiring the installation of QGIS.
 - **QGIS**, especially its Python interface and API **PyQGIS**, has been extensively used throughout this project. However, since it requires defining the application path to access its algorithms, all projects have been containerized with **Docker**. Its QGIS image allowed to enhance reusability and independence.
 - To make the final user aware of the functions available a **Read the docs** documentation have been developed [<https://talea-documentation.netlify.app/>]

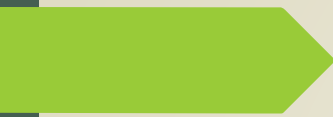




Data collection

- The data have been collected from [Open Data Bologna](#) and [Geoportale Emilia Romagna](#)
- Four different typologies of data can be individuated:
 - *General information about Bologna land usage*
 - *Green dataset*
 - *Human infrastructure dataset*
 - *Statistical dataset*





General information about Bologna land usage

- ▶ The dataset "**uso del suolo**" constitutes the mapping of the various uses of the territory
- ▶ In this study the database has been used to discriminate the areas marked as available to build green space on top of them

Statistical dataset

- ▶ **Aree statistiche:** contains the division of the municipal territory into 90 statistical areas;
- ▶ **Popolazione per area statistica:** contains the population of Bologna, per each statistical area.

Green dataset

- **Verde in gestione:** Management units (green areas) present in the municipal territory of the City of Bologna;
- **Verde Privato:** mapping of green areas within the urbanized territory, excluding public green spaces under maintenance;
- **Alberi In Manutenzione:** coordinates of all the trees planted in the city of Bologna;
- **Aree Boschive** and **Aree Fluviali:** area of land covered by vegetation and by water.



Human infrastructure dataset

- **Edifici particellari:** contains the list of buildings with parcel details extracted from the Municipal Technical Map;
- **Aree stradali:** street areas in the city of Bologna.





Where should we build new green areas?

- Using a degree of approximation:
 - **Free space**
 - **Road**
- *Free space* is intended as the remaining portion of the city once buildings, roads, railways and the green space is removed.
- Yard court is included in the free space.
- Roads, in a small percentage, can be designed as places to insert new green, such as trees.

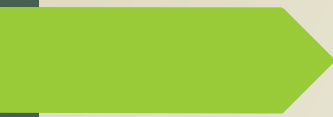


Data Preprocessing

- Heavy preprocessing has been required due to:
 - Lack of a usable structure
 - Presence of duplicates
 - Geospatial information were corrupted
 - Presence of missing data
- QGIS is one of the most common tool to work with geospatial data, offering also a python interface

Data Preprocessing: New dataset

- ▶ Creation of a **unified green dataset** by combining all existing green datasets, avoiding duplicates.
- ▶ Fixed corrupted geospatial information contained inside "un_gest".
- ▶ The road dataset contains squares and highways that needs to be removed.
 - **No existing dataset for squares**, the attribute '*Tronco di intersezione tra strade a raso*' has been properly filtered to delete squares;
 - **Highways** have been filtered using the attribute '*Autostrade e superstrade*' in the dataset "uso_del_suolo".
- ▶ **Free space computation** have been delimited to some areas of the map corresponding to the features '*Tessuto residenziale compatto e denso*', '*Tessuto residenziale urbano*', '*Tessuto residenziale rado*' contained in the dataset "uso_del_suolo".

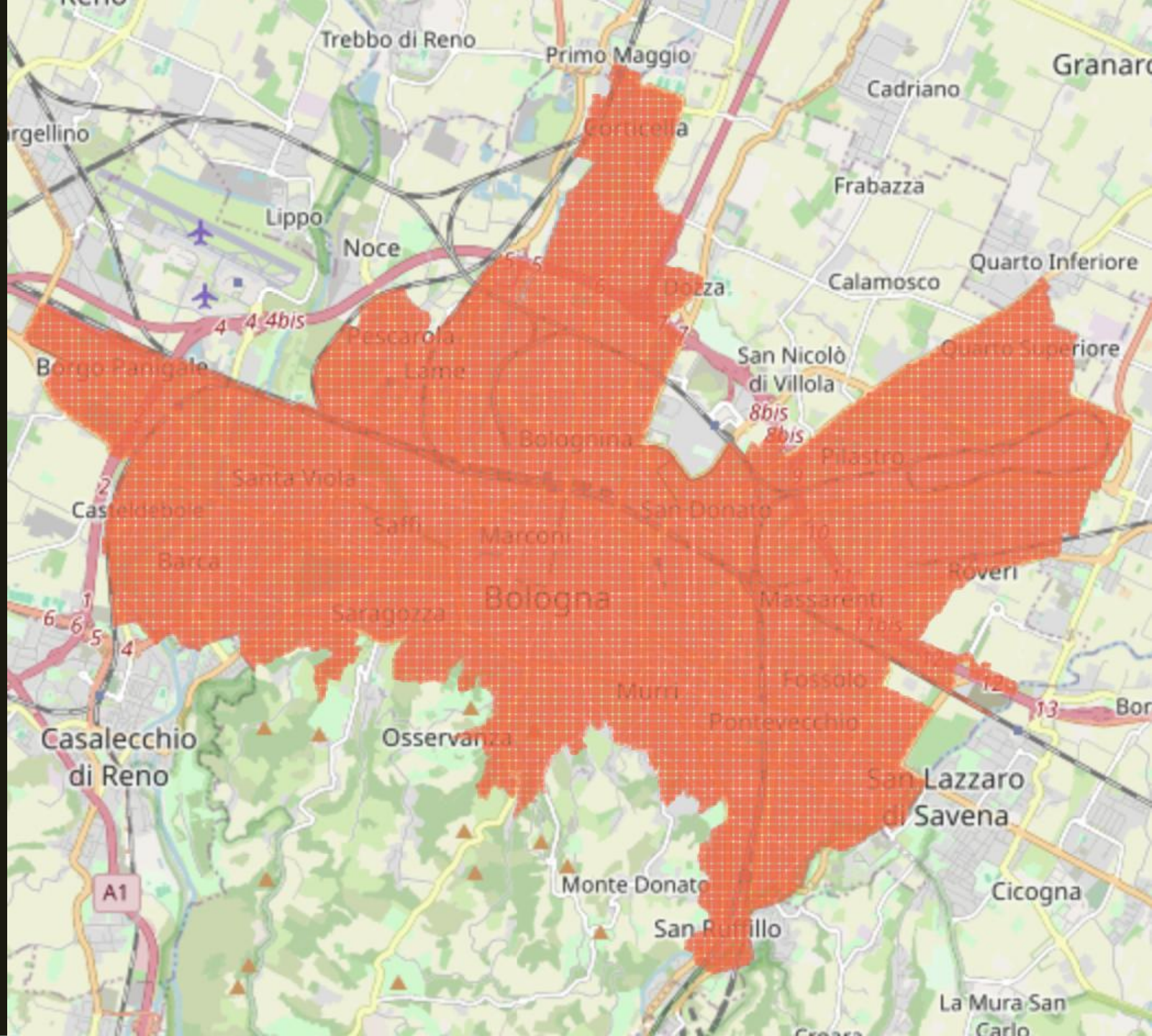


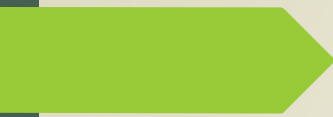
Data Preprocessing: Boundary

- The boundary of Bologna have been taken from the dataset "*aree statistiche*".
- Some areas have been excluded to avoid biased results:
 - 'Colli' have been considered as a complete green area and no new green cells are necessary.
 - 'Rigosa', 'Aereoporto', 'Savena Abbandonato', 'Fiera' have been excluded, due to missing buildings data and green areas.

Data Preprocessing: Final dataset

- ▶ The final dataset is a grid made of 100x100 meters cells
- ▶ The **cells features** are:
 - Area of the green space;
 - Number of trees;
 - Area of free spaces;
 - Number of free space areas (highlight fragmentation);
 - Area of roads





Data Preprocessing: Macro-scale aggregation

- ▶ Leveraging on the areas division in "*aree_statistiche*", two new features have been computed:
 - ***Population density***
 - ***Macro-level green areas***
- ▶ Population density is computed as the information inside "*popolazione-per-area-statistica*" dataset divided by the area of the statistical area itself.
- ▶ Macro-level green areas aggregates the amount of green inside each statistical area.



Model

- The problem has been formalized using **Minizinc** and run using the Mixed Integer Programming (MIP) solver **HiGHS**, ensuring good performance because of two reasons:
 - *Small problem instance*
 - *Lightweight model*
- The model aims to find the best n areas to place a Talea Green Cell (TGC), rated according to a **utility function**.
- The **utility function** is composed of two factors:
 - *macro factor*
 - *micro factor*

Model: Utility function

- For each cell i_{th} it is defined as:

$$utility_i = macro_factor_i \times \frac{\alpha_{yard} \cdot yard_cells_i + \beta_{streets} \cdot street_cells_i}{num_areas_i + green_space_i + num_trees_i}$$

- macro factor

Multiplicative factor referred to the whole statistical area

$$macro_factor_i = density_param \cdot density_factor_i + green_param \cdot green_factor_i$$

where

$$density_param, green_param \in [0 .. 1]$$

- The remaining part of the formulation represents the micro factor, referred to the grid cells level and is the core utility function.

Model: Utility function

► street cells

Road area covered by a TGC, whose influence on the utility is governed by the parameter $\beta_{streets} \in [0..1]$

$$street_cells_i \leq street_area_i$$

► yard cells

Area of the building's yards covered by a TGC, whose influence on the utility is governed by the parameter $\alpha_{yards} \in [0..1]$

$$yards_cells_i \leq free_space_i$$

► num areas Number of yard areas inside a cell, for the contiguity

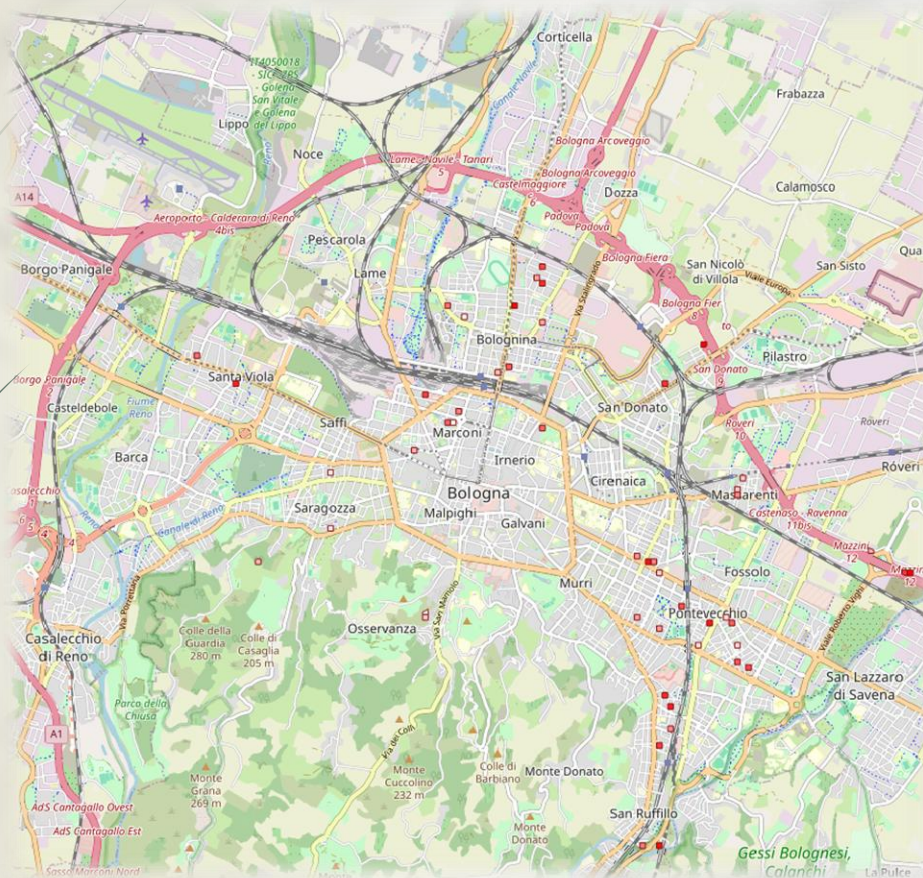
► green space Amount of green area already present in a cell

► num trees Number of trees inside a cell

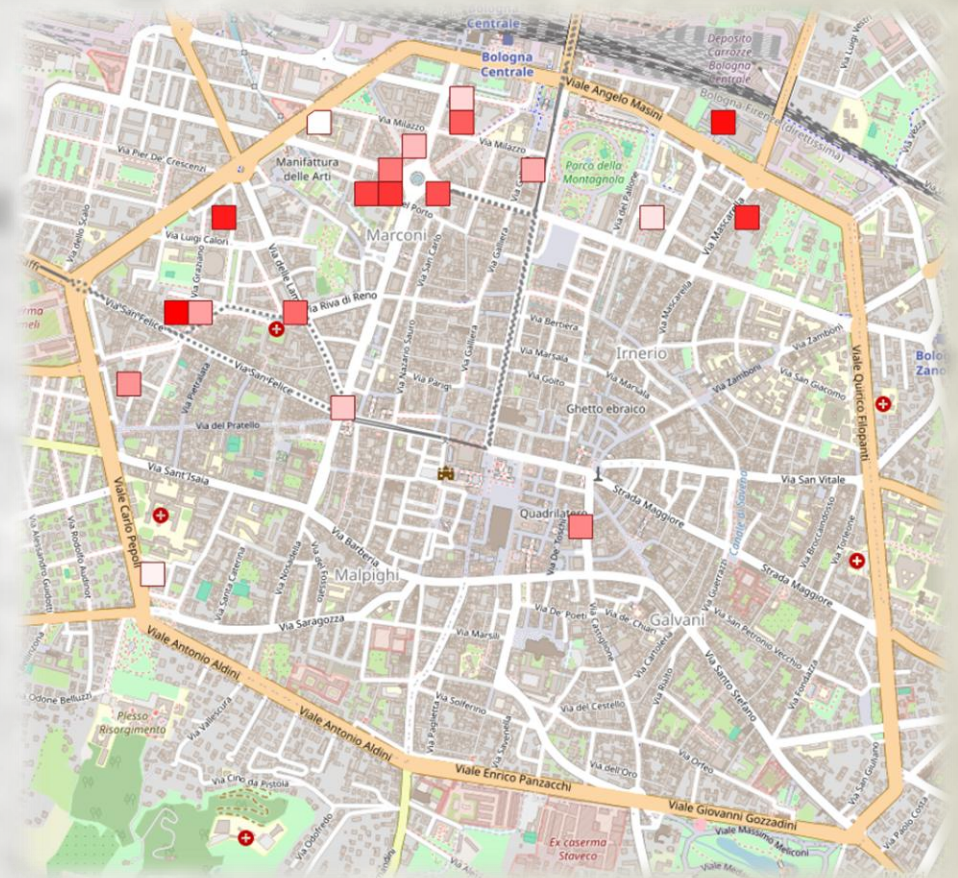
Results

- ▶ The optimization model developed has been applied to *two different spatial scales*:
 - ▶ **City center**
 - ▶ **Entire municipal area**
- ▶ For each case study, different configurations of the previously described parameters have been tested:
 - ▶ number of cells
 - ▶ *density_param*
 - ▶ *green_param*
 - ▶ *streets_param* ($\beta_{streets}$)
 - ▶ *yard_param* (α_{yards})

Results



Entire Bologna municipality - default configuration



City center - default configuration



What are the main insights?

- It has been compared the previous utility with the following:

$$utility_i = macro_factor_i \times$$

$$(\alpha_{yard} \cdot yard_cells_i + \beta_{streets} \cdot street_cells_i - num_areas_i - green_space_i - num_trees_i)$$

- The *macro factor* decides which statistical area to enhance.
- The "*micro*" components subtracted are not capable of influencing the placement.
- The "*micro*" components only influence the utility values.

What are the main insights?

► In the center:

- Areas that are identified by architectural experts as ideal for the TGCs placement have been successfully detected by our model, like:
 - *Martiri – Boldrini*
 - *Via Milazzo*
 - *Porta Mascarella*
- No big changes among one parametrization and the other, apart from the “no yard” solution

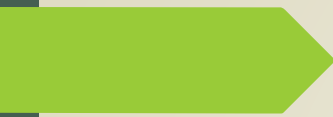


- entrances to public parks ●
- green areas ●
- spaces/buildings of interest (squares/schools/museums/etc.) ●
- ongoing projects (buildings/spaces/infrastructure/etc.) ●



What are the main insights?

- ▶ In the entire Bologna municipality:
 - Valuable areas of the center are in part detected also when including the suburbs
 - The higher need of new green spaces is shown by areas like:
 - *The center*
 - *Pontevecchio*
 - *Bolognina*
 - *Saragozza*
 - Boosting the green parameter w.r.t. the density one leads to a higher focus on industrial areas and suburbs
 - Boosting the street or yard placement leads to inverse distributions of new TGCs



Future Works

- **Expanding the geographical scope** with all the city areas that have been omitted due to insufficient or outdated information.
- Integrating additional variables for a more accurate urban development scenario, like **accessibility to public transport nodes, micro-climate data, heat island effects**, etc.
- Including **technical or legal constraints on construction sites** – related to the challenges of building on existing road infrastructure or bureaucratic complexities.
- On the technical side, **extending the GUI** for a more data-driven approach in the decision-support framework with:
 - Real-time computation
 - Dynamic scenario simulations
 - Advanced visualization tools

A solid green arrow pointing to the right, positioned on the left side of the slide.

THANK YOU FOR THE ATTENTION!

Christian Di Buò
Simone Reale